

Capstone Project:

Bluestock Mutual Fund Analytics

Executive Summary

The Bluestock Mutual Fund Analytics Capstone Project was designed to build a comprehensive data analytics and business intelligence solution for analyzing the Indian mutual fund industry. The project integrates data engineering, database management, exploratory data analysis, financial performance evaluation, advanced risk analytics, and interactive dashboard development into a single end-to-end platform. The primary objective was to transform raw mutual fund data into meaningful insights that can support investment decision-making, portfolio evaluation, and market trend analysis.

The project utilized multiple datasets covering fund master information, historical NAV data, assets under management (AUM), monthly SIP inflows, category-wise inflows, industry folio counts, scheme performance metrics, investor transactions, portfolio holdings, and benchmark indices. These datasets were collected, cleaned, validated, and transformed through a structured ETL (Extract, Transform, Load) process using Python and Pandas. The cleaned data was then stored in a SQLite database designed using a star-schema approach to improve data organization and analytical efficiency.

Exploratory Data Analysis (EDA) was performed to understand industry trends, investor behavior, and fund performance. Multiple visualizations were created to analyze NAV movements, AUM growth, SIP inflow trends, investor demographics, geographic distribution of investments, category preferences, and market participation patterns. These analyses helped identify key industry trends and provided valuable insights into how investors interact with mutual fund products.

The project further focused on performance analytics by calculating important financial metrics such as daily returns, Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown. These metrics were used to evaluate fund performance, measure risk-adjusted returns, assess benchmark sensitivity, and identify the best-performing schemes across different categories. Advanced analytics techniques including Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratios, Investor Cohort Analysis, SIP Continuity Analysis, and Sector Concentration Analysis were also implemented to provide deeper insights into risk exposure and investor engagement.

To make the findings accessible to business users and decision-makers, an interactive Power BI dashboard was developed. The dashboard consists of four analytical pages covering Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends. Through interactive

charts, KPI cards, filters, and drill-down capabilities, users can explore fund performance, investor behavior, market trends, and risk metrics in a user-friendly environment.

The project successfully demonstrates the application of modern data analytics techniques in the financial services domain. It provides a scalable framework for mutual fund analysis and highlights the value of combining data engineering, statistical analysis, financial modeling, and business intelligence tools. The final solution delivers actionable insights for investors, analysts, and fund managers while establishing a strong foundation for future enhancements such as real-time data integration, predictive analytics, portfolio optimization, and automated reporting systems.

1. Introduction

Problem Statement

The mutual fund industry generates large amounts of data, including NAV history, investor transactions, portfolio holdings, benchmark performance, and SIP inflows. Analyzing this data manually is difficult and time-consuming. Therefore, there is a need for an integrated analytics platform that can clean, process, and analyze mutual fund data to generate meaningful insights on fund performance, investor behavior, risk, and market trends. This project aims to build such a platform using Python, SQL, SQLite, and Power BI to support data-driven investment decisions.

Objectives

1. Build an ETL Pipeline

To develop an automated Extract, Transform, and Load (ETL) pipeline that collects raw mutual fund data, processes it efficiently, and prepares it for analysis. This reduces manual effort and ensures a consistent data flow throughout the project.

2. Clean and Validate Data

To improve the quality and reliability of the datasets by handling missing values, removing duplicate records, correcting data formats, and validating important fields. High-quality data is essential for accurate analysis and reporting.

3. Create a SQLite Database

To design and implement a centralized SQLite database for storing cleaned mutual fund data. The database enables efficient querying, better data management, and seamless integration with analytical tools.

4. Perform Exploratory Data Analysis (EDA)

To explore and understand mutual fund data through statistical analysis and visualizations. This

helps identify trends, patterns, investor behavior, and market movements that may not be visible from raw data alone.

5. Calculate Fund Performance Metrics

To evaluate mutual fund performance using key financial metrics such as Daily Returns, CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown. These metrics provide insights into both returns and risk-adjusted performance.

6. Develop an Interactive Power BI Dashboard

To create a user-friendly dashboard that presents important insights through charts, KPI cards, tables, and filters. The dashboard allows users to interact with the data and gain meaningful insights quickly.

7. Generate Advanced Risk Analytics

To assess investment risk using advanced analytical techniques such as Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, and Sector Concentration Analysis. These measures help understand potential losses and portfolio stability.

8. Provide Actionable Insights and Recommendations

To transform complex financial data into meaningful insights that can support investors, analysts, and fund managers in making informed investment decisions. The project also includes a simple recommendation system based on risk profiles and fund performance.

2. Data Sources

This project uses ten different datasets that provide comprehensive information about the mutual fund industry. These datasets cover fund details, historical performance, investor activity, industry growth, portfolio composition, and benchmark market data. Together, they form the foundation for analysis, reporting, and dashboard development.

1. Fund Master Data

This dataset contains basic information about all mutual fund schemes, including scheme names, AMFI codes, fund categories, fund houses, investment plans, and risk classifications. It serves as the primary reference table for identifying and categorizing mutual funds throughout the project.

2. NAV History

The NAV (Net Asset Value) History dataset contains daily NAV values for different mutual fund schemes over a period of time. It is used to analyze fund performance, calculate daily returns, identify trends, and compute important financial metrics such as CAGR, Sharpe Ratio, and Maximum Drawdown.

3. AUM by Fund House

This dataset provides information on Assets Under Management (AUM) for various fund houses. AUM represents the total market value of assets managed by an asset management company (AMC). It helps evaluate the size, growth, and market share of different fund houses within the mutual fund industry.

4. Monthly SIP Inflows

This dataset contains monthly Systematic Investment Plan (SIP) inflow data. SIP inflows indicate the amount of money invested regularly by investors into mutual funds. The dataset is used to analyze investment trends, investor confidence, and the growth of systematic investing over time.

5. Category Inflows

The Category Inflows dataset records net inflows and outflows across different mutual fund categories such as equity, debt, hybrid, and index funds. It helps identify investor preferences and understand which categories attract the highest levels of investment.

6. Industry Folio Count

This dataset tracks the number of folios in the mutual fund industry. A folio represents an individual investor account. The data is useful for measuring industry participation, monitoring investor growth, and evaluating the expansion of the mutual fund sector.

7. Scheme Performance

The Scheme Performance dataset contains performance-related information for mutual fund schemes, including returns, expense ratios, risk grades, and other investment metrics. It is used to compare fund performance and evaluate risk-adjusted returns across different schemes.

8. Investor Transactions

This dataset records individual investor transactions such as SIP investments, lump-sum purchases, redemptions, and switches. It provides insights into investor behavior, transaction patterns, investment preferences, demographic characteristics, and geographic distribution.

9. Portfolio Holdings

The Portfolio Holdings dataset contains details of securities held by mutual funds, including sector allocations and portfolio weights. It is used to analyze diversification, sector concentration, and portfolio risk through measures such as the Herfindahl-Hirschman Index (HHI).

10. Benchmark Indices

This dataset includes historical values of benchmark market indices such as Nifty 50 and Nifty 100. Benchmark data is used to compare mutual fund performance against the broader market and calculate performance measures such as Alpha, Beta, and Tracking Error.

3. ETL Design

The ETL (Extract, Transform, Load) process is a crucial part of this project as it ensures that raw mutual fund data is converted into a clean, structured, and analysis-ready format. The ETL pipeline was developed using Python and Pandas to automate data processing and improve data quality.

Data Ingestion

The first step of the ETL process was data ingestion, where raw datasets were collected and imported into the Python environment. All datasets were available in CSV format and were loaded using the Pandas library. This step allowed the data to be accessed, inspected, and prepared for further processing.

The imported datasets included fund master information, NAV history, AUM data, SIP inflows, category inflows, folio counts, scheme performance details, investor transactions, portfolio holdings, and benchmark indices. Loading all datasets into a common environment ensured consistency and enabled efficient data manipulation.

Data Cleaning

After ingestion, the raw data underwent a comprehensive cleaning process to improve its accuracy and reliability. Data cleaning is important because raw datasets often contain missing values, inconsistent formats, duplicate records, and incorrect data types that can affect analysis results.

The following cleaning activities were performed:

Missing Value Handling:

Missing or null values were identified and handled appropriately using techniques such as filling, replacing, or removing incomplete records depending on the nature of the data.

Duplicate Removal:

Duplicate records were detected and removed to prevent double-counting and ensure the accuracy of calculations and reporting.

Date Standardization:

Date columns from different datasets were converted into a consistent format to support time-series analysis and proper data integration.

Numeric Conversion:

Columns containing financial values such as NAV, AUM, SIP amounts, returns, and portfolio weights were converted into appropriate numeric formats for calculations and statistical analysis.

Column Validation:

Column names, data types, and business rules were validated to ensure consistency across all datasets. This helped prevent errors during database loading and analytical processing.

Through these cleaning steps, the datasets became more reliable, consistent, and suitable for advanced analysis.

Data Storage

Once the data was cleaned and validated, it was loaded into a SQLite database for centralized storage and management. SQLite was selected because it is lightweight, easy to use, and supports efficient querying without requiring a separate database server.

The cleaned datasets were organized into structured tables within the database, enabling faster data retrieval and analysis. Storing the data in SQLite also simplified integration with analytical workflows and reporting tools. The database served as the primary source for SQL queries, performance analysis, and dashboard development throughout the project.

The ETL process ensured that all data used in the project was accurate, organized, and ready for further analysis, forming the foundation for the EDA, performance analytics, risk assessment, and Power BI dashboard components of the project.

4. Database Design

A well-structured database design is essential for efficient data storage, retrieval, and analysis. In this project, a dimensional data model was used to organize the mutual fund datasets into dimension tables and fact tables. This approach helps improve query performance, reduce data duplication, and simplify reporting and dashboard development.

The database was implemented using SQLite and follows a star-schema-like structure, where dimension tables provide descriptive information and fact tables store transactional and measurable data.

Dimension Tables

Dimension tables contain descriptive attributes that provide context to the data stored in fact tables.

1. dim_fund

The **dim_fund** table contains information about mutual fund schemes, including fund names, AMFI codes, fund houses, categories, plans, and risk grades. This table serves as a reference table that helps identify and classify different mutual funds across the database.

Purpose:

- Store scheme-related information.
- Support fund-level analysis and reporting.
- Enable filtering and grouping by fund category, AMC, or risk level.

2. dim_date

The **dim_date** table contains date-related attributes such as day, month, quarter, and year. It provides a standardized calendar structure that supports time-series analysis and trend reporting.

Purpose:

- Enable monthly, quarterly, and yearly analysis.
- Support trend visualization and time-based calculations.
- Improve consistency across different datasets.

Fact Tables

Fact tables store measurable and transactional data that can be analyzed using dimension tables.

1. fact_nav

The **fact_nav** table stores daily Net Asset Value (NAV) records for mutual fund schemes. It contains information such as fund identifiers, dates, and NAV values.

Purpose:

- Track NAV movements over time.
- Calculate daily returns and performance metrics.
- Support trend and benchmark analysis.

2. fact_transactions

The **fact_transactions** table stores investor transaction records, including SIP investments, lump-sum purchases, redemptions, transaction dates, and investment amounts.

Purpose:

- Analyze investor behavior.
- Study transaction patterns and SIP continuity.
- Generate demographic and geographic insights.
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3. fact_performance

The **fact_performance** table contains scheme-level performance metrics such as returns, expense ratios, risk grades, and other financial indicators.

Purpose:

- Compare mutual fund performance.
- Calculate risk-adjusted return metrics.
- Support fund ranking and recommendation systems.

4. fact_aum

The **fact_aum** table stores Assets Under Management (AUM) information for different fund houses and schemes.

Purpose:

- Measure industry growth.
- Compare market share among AMCs.
- Analyze fund house performance over time.

5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the underlying patterns, trends, and relationships within the mutual fund datasets. Various statistical techniques and visualizations were used to examine fund performance, investor behavior, industry growth, and market trends. The insights obtained from EDA provided the foundation for further performance analytics and dashboard development.

NAV Trend Analysis

The Net Asset Value (NAV) history of all mutual fund schemes was analyzed over the study period to understand long-term performance and market movements. Line charts were created to visualize daily NAV fluctuations and identify periods of growth, correction, and recovery.

Key Observation:

Most mutual fund schemes demonstrated a steady upward trend in NAV over the long term, indicating positive wealth creation for investors. Although temporary market corrections were observed during certain periods, the overall growth trajectory remained strong, highlighting the resilience of mutual fund investments.

AUM Analysis

Assets Under Management (AUM) data was analyzed to compare the size and market presence of different Asset Management Companies (AMCs). Bar charts and comparative visualizations were used to identify leading fund houses based on total assets managed.

Key Observation:

SBI Mutual Fund maintained the highest AUM contribution among all fund houses, indicating its strong market position and large investor base. The analysis also revealed significant differences in market share across AMCs, with a few major players controlling a substantial portion of industry assets.

SIP Trends Analysis

Monthly Systematic Investment Plan (SIP) inflow data was examined to understand investor participation and investment behavior over time. Time-series visualizations were used to track changes in SIP contributions between 2022 and 2025.

Key Observation:

SIP inflows showed strong and consistent growth throughout the analysis period. The increasing trend indicates growing investor confidence in mutual funds and reflects the popularity of disciplined, long-term investing through SIPs. Record-high SIP inflows towards the end of the study period further demonstrated the expanding reach of mutual fund investments.

Investor Demographics Analysis

Investor demographic data was analyzed to understand the distribution of investors across different age groups and genders. Pie charts, bar charts, and box plots were used to examine participation levels and investment behavior.

Key Observation:

The majority of investors belonged to the 25–40 age group, making it the most active investor segment. This suggests that young and middle-aged individuals are increasingly adopting mutual funds as a preferred investment vehicle for long-term financial planning and wealth creation.

Geographic Analysis

Geographic data was analyzed to study the distribution of mutual fund investments across different states and city tiers. Visualizations were created to compare transaction volumes and investment activity by location.

Key Observation:

Tier-1 cities contributed the largest share of transaction volumes and investment amounts. These urban centers demonstrated higher participation levels due to greater financial awareness, better

access to investment platforms, and higher disposable incomes. However, growing participation from smaller cities indicates the expanding penetration of mutual funds across the country.

6. Performance Analytics

Performance Analytics was conducted to evaluate the returns, risk, and overall effectiveness of mutual fund schemes. Various financial metrics were calculated using historical NAV data and benchmark indices to assess both absolute and risk-adjusted performance. These metrics provide a comprehensive understanding of how funds have performed over time and how they compare with market benchmarks.

Daily Returns

Daily returns were calculated using the percentage change in Net Asset Value (NAV) from one trading day to the next. Daily returns form the foundation for most financial performance and risk calculations, as they reflect the short-term movement of a fund's value.

Purpose:

- Measure day-to-day fund performance.
- Analyze return distribution and volatility.
- Serve as input for Sharpe Ratio, Sortino Ratio, Alpha, Beta, and VaR calculations.

Key Observation:

Most funds showed moderate daily fluctuations, with returns generally centered around positive long-term growth trends.

Compound Annual Growth Rate (CAGR)

CAGR was calculated for 1-year, 3-year, and 5-year investment horizons. It represents the annualized growth rate of an investment over a specific period, assuming profits are reinvested.

Purpose:

- Measure long-term investment growth.
- Compare performance across different funds.
- Identify consistently growing schemes.

Key Observation:

Several equity-oriented funds achieved strong CAGR values over longer periods, highlighting their ability to generate sustainable returns.

Sharpe Ratio

The Sharpe Ratio was used to evaluate risk-adjusted returns by comparing a fund's excess return over the risk-free rate with its overall volatility.

Purpose:

- Measure return generated per unit of risk.
- Compare funds with different risk levels.
- Identify efficient investment opportunities.

Key Observation:

Funds with higher Sharpe Ratios delivered superior returns relative to the amount of risk taken, making them attractive options for investors seeking balanced performance.

Sortino Ratio

The Sortino Ratio is similar to the Sharpe Ratio but focuses only on downside volatility. It penalizes harmful negative fluctuations while ignoring positive return variations.

Purpose:

- Evaluate downside risk-adjusted performance.
- Measure how effectively a fund manages losses.
- Identify funds with stable return patterns.

Key Observation:

Funds with high Sortino Ratios demonstrated better protection against downside risk while maintaining competitive returns.

Alpha and Beta

Alpha and Beta were calculated using benchmark index returns to assess fund performance relative to the market.

Alpha measures the excess return generated by a fund beyond what would be expected based on market movements.

Beta measures the sensitivity of a fund's returns to changes in the benchmark index.

Purpose:

- Evaluate fund manager effectiveness.
- Measure market-related risk.
- Compare fund performance against benchmark indices.

Key Observation:

Positive Alpha values indicated that certain funds outperformed their benchmarks, while Beta values revealed varying levels of market sensitivity across schemes.

Maximum Drawdown

Maximum Drawdown was calculated to determine the largest percentage decline from a historical peak to a subsequent trough. This metric highlights the worst loss an investor could have experienced during the analysis period.

Purpose:

- Measure downside risk.
- Assess potential investment losses.
- Compare fund resilience during market downturns.

Key Observation:

High-risk funds generally experienced larger drawdowns during market corrections, while diversified funds showed greater resilience and faster recovery.

7. Advanced Analytics

In addition to traditional performance analysis, advanced analytical techniques were applied to gain deeper insights into risk exposure, investor behavior, portfolio diversification, and fund selection. These methods help investors and fund managers make more informed decisions by evaluating potential risks and identifying investment opportunities beyond standard return metrics.

Historical Value at Risk (VaR)

Historical Value at Risk (VaR) at the 95% confidence level was calculated for all mutual fund schemes using historical daily return data. VaR estimates the maximum expected loss that may occur under normal market conditions over a given period.

Purpose:

- Measure downside risk exposure.
- Estimate potential losses in adverse market conditions.
- Compare risk levels across different funds.

Key Observation:

High-risk equity funds generally exhibited larger VaR values, indicating a greater probability of significant short-term losses compared to conservative funds.

Conditional Value at Risk (CVaR)

Conditional Value at Risk (CVaR), also known as Expected Shortfall, was calculated as the average loss beyond the VaR threshold. While VaR identifies the loss limit, CVaR measures the severity of losses when that limit is exceeded.

Purpose:

- Evaluate extreme downside risk.
- Assess potential losses during market stress.
- Provide a more comprehensive risk measure than VaR alone.

Key Observation:

Funds with higher CVaR values were more vulnerable to extreme market downturns, highlighting the importance of risk management and diversification.

Rolling Sharpe Ratio

A 90-day Rolling Sharpe Ratio was calculated to evaluate the consistency of risk-adjusted performance over time. Unlike a single Sharpe Ratio value, the rolling metric captures changes in performance across different market conditions.

Purpose:

- Monitor performance stability.
- Identify periods of strong or weak risk-adjusted returns.
- Compare consistency among funds.

Key Observation:

Top-performing funds maintained relatively stable rolling Sharpe Ratios, demonstrating consistent performance despite market fluctuations.

Investor Cohort Analysis

Investors were grouped into cohorts based on the year of their first investment transaction. Metrics such as average SIP amount, total investment value, and preferred fund categories were analyzed for each cohort.

Purpose:

- Understand long-term investor behavior.

- Compare investment patterns across generations of investors.
- Identify high-value investor segments.

Key Observation:

Recent investor cohorts showed higher participation in equity-oriented funds and larger average SIP contributions, reflecting growing confidence in long-term wealth creation through mutual funds.

SIP Continuity Analysis

SIP continuity was analyzed by measuring the gap between consecutive SIP transactions. Investors with an average gap exceeding 35 days were classified as "at-risk" investors.

Purpose:

- Monitor investor engagement.
- Identify potential SIP discontinuation risks.
- Support investor retention strategies.

Key Observation:

Most investors maintained regular SIP schedules, while a smaller segment displayed irregular contributions, indicating a higher likelihood of future discontinuation.

Sector Concentration Analysis (HHI)

The Herfindahl-Hirschman Index (HHI) was calculated using portfolio holding weights to measure sector concentration and diversification. Higher HHI values indicate more concentrated portfolios, while lower values suggest greater diversification.

Purpose:

- Evaluate portfolio diversification.
- Measure concentration risk.
- Compare investment strategies across funds.

Key Observation:

Some equity funds exhibited high sector concentration, increasing exposure to sector-specific risks, whereas diversified funds demonstrated lower concentration and better risk distribution.

Fund Recommendation Engine

A simple recommendation engine was developed to suggest mutual funds based on an investor's risk appetite (Low, Moderate, or High). The system ranked funds using Sharpe Ratio and risk-grade classifications.

Purpose:

- Assist investors in fund selection.
- Provide personalized investment recommendations.
- Demonstrate practical application of analytics results.

Key Observation:

The recommendation engine successfully identified top-performing funds within each risk category, enabling users to select investments aligned with their financial goals and risk tolerance.

8. Dashboard Overview

An interactive Power BI dashboard was developed to present key insights from the mutual fund analytics project in a visually appealing and user-friendly format. The dashboard enables users to explore industry trends, fund performance, investor behavior, and market dynamics through interactive charts, KPIs, filters, and drill-down capabilities. It consists of four dedicated pages, each focusing on a specific analytical area.

Page 1 – Industry Overview

The Industry Overview page provides a high-level summary of the mutual fund industry and its overall growth. It helps users quickly understand the scale and performance of the industry through key performance indicators and trend visualizations.

Key Components:

- **Total AUM (Assets Under Management):** Displays the overall value of assets managed by mutual funds.
- **Total SIP Inflows:** Shows the cumulative SIP investments received during the analysis period.
- **Total Folios:** Represents the total number of investor accounts in the mutual fund industry.
- **Total Schemes:** Displays the total number of mutual fund schemes available for investment.
- **AUM Trend Analysis:** Visualizes industry growth over time through line charts.
- **AUM by Fund House:** Compares the market share of different Asset Management Companies (AMCs).

Page 2 – Fund Performance

The Fund Performance page focuses on evaluating and comparing mutual fund schemes using performance and risk metrics. It enables users to identify high-performing funds and assess their risk-return characteristics.

Key Components:

- **Risk vs Return Scatter Plot:** Compares funds based on returns and volatility, helping identify efficient investment options.
- **Fund Scorecard Table:** Displays important performance indicators such as CAGR, Sharpe Ratio, Alpha, Beta, and overall fund scores.
- **Benchmark Comparison Chart:** Compares fund performance against benchmark indices such as Nifty 50 and Nifty 100.
- **Interactive Filters:** Allows users to filter data by fund house, category, and investment plan.

Page 3 – Investor Analytics

The Investor Analytics page provides insights into investor demographics, geographic distribution, and transaction behavior. It helps understand who is investing and how investment patterns vary across regions and investor groups.

Key Components:

- **State-wise Investment Analysis:** Displays transaction amounts across different states.
- **Transaction Distribution:** Shows the proportion of SIPs, lump-sum investments, and redemptions using donut charts.
- **Age Group Analysis:** Examines investment behavior across different age categories.
- **Monthly Transaction Trends:** Tracks changes in transaction volumes over time.
- **Interactive Slicers:** Enable filtering by state, age group, and city tier.

Page 4 – SIP & Market Trends

The SIP & Market Trends page explores the relationship between systematic investments and broader market performance. It also highlights category-wise investment flows and investor preferences.

Key Components:

- **SIP vs Nifty Trend:** Compares SIP inflows with benchmark market movements over time.
- **Category Inflow Heatmap:** Visualizes investment activity across different mutual fund categories and periods.

- **Top Categories by Net Inflow:** Identifies the categories attracting the highest investor interest.
- **Trend Analysis Charts:** Show changes in investor preferences and market sentiment.

9.Key Findings

The analysis of mutual fund performance, investor behavior, and industry trends revealed several important insights that can support better investment decision-making and portfolio management.

1. SIP Inflows Showed Consistent Growth

Monthly SIP inflows increased steadily throughout the study period, indicating growing investor confidence in mutual funds and the increasing popularity of disciplined long-term investing.

2. Equity Funds Were the Most Preferred Category

Among all mutual fund categories, equity funds attracted the highest investor participation and inflows, reflecting investors' preference for long-term wealth creation and capital appreciation.

3. SBI Mutual Fund Maintained the Highest AUM

SBI Mutual Fund consistently recorded the largest Assets Under Management (AUM), highlighting its strong market presence, investor trust, and extensive product portfolio.

4. Investors Aged 25–40 Dominated Investment Activity

Demographic analysis revealed that the 25–40 age group accounted for the majority of transactions and investments, making it the most active segment in the mutual fund industry.

5. Funds with Higher Sharpe Ratios Delivered Better Risk-Adjusted Returns

Performance analysis showed that funds with strong Sharpe Ratios generally provided higher returns relative to the level of risk taken, making them attractive investment options.

6. Several Funds Displayed High Sector Concentration

Portfolio holding analysis indicated that some mutual funds were heavily concentrated in specific sectors, increasing their exposure to sector-specific risks and market fluctuations.

7. High-Risk Funds Experienced Larger Drawdowns

Maximum Drawdown analysis revealed that aggressive and high-risk funds suffered greater losses during market downturns compared to more diversified and balanced funds.

10. Limitations

- The analysis is based only on **historical data**, and past performance may not guarantee future results.
- Benchmark analysis was limited by the availability and quality of benchmark index data.
- The project does not include **real-time NAV or live market data integration**.
- The fund recommendation engine uses a **simplified logic** based mainly on risk grade and Sharpe Ratio.
- Investor analysis was restricted to the information available in the dataset and did not include factors such as income, occupation, or financial goals.
- Portfolio concentration analysis was performed using a **snapshot of holdings data**, which may change over time.
- The project focused only on the provided datasets and did not include external economic indicators such as inflation, interest rates, or GDP data.

12. Conclusion

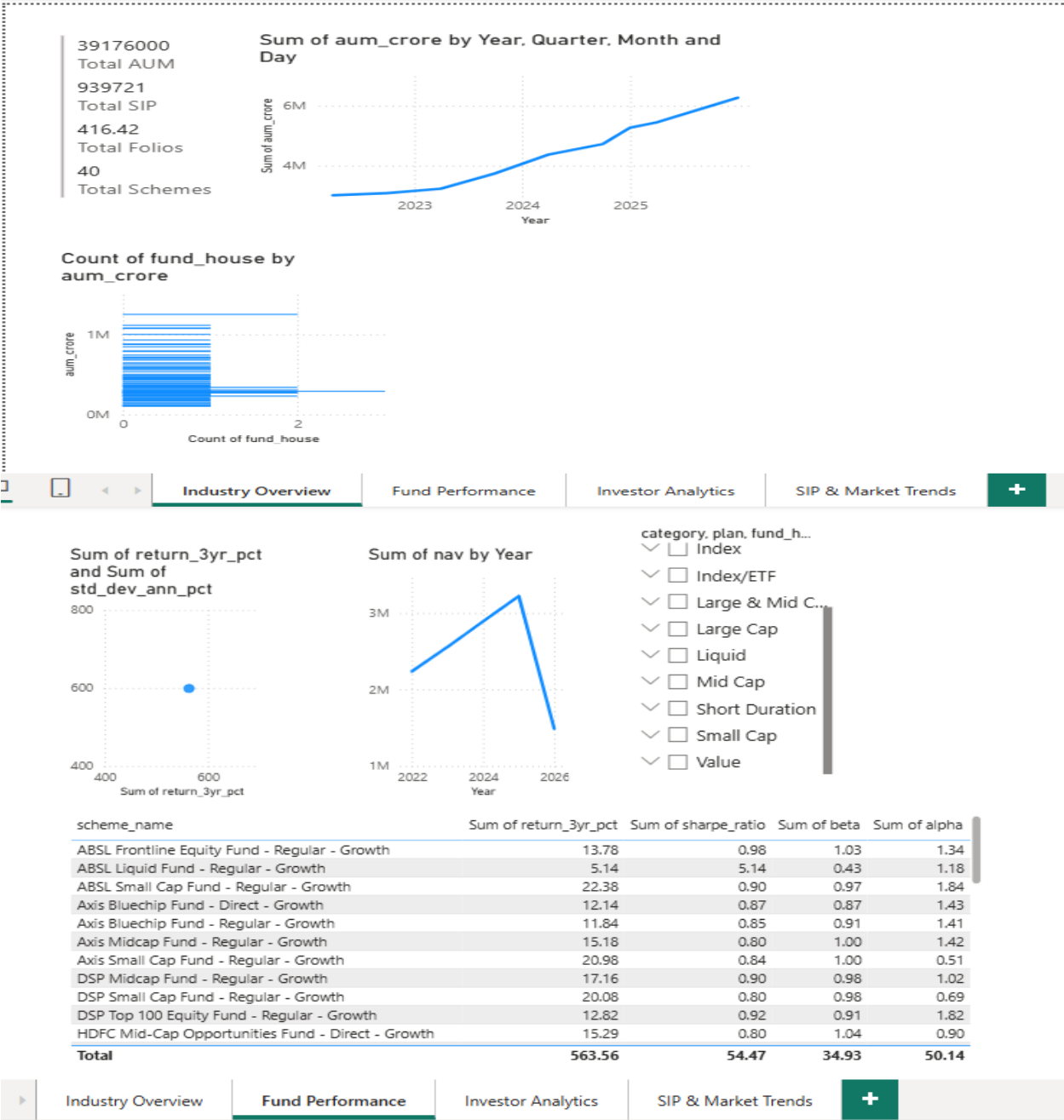
The Bluestock Mutual Fund Analytics Capstone Project successfully developed a comprehensive analytics platform for evaluating mutual fund performance, investor behavior, industry trends, and portfolio risk. The project covered the complete data analytics lifecycle, beginning with data ingestion and cleaning, followed by database creation, exploratory data analysis, performance evaluation, advanced analytics, and dashboard development.

Using Python, Pandas, SQLite, SQL, and Power BI, multiple datasets were transformed into a structured and analysis-ready format. Key performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown were calculated to assess fund performance and risk-adjusted returns. Advanced analytics techniques including VaR, CVaR, Rolling Sharpe Ratio, Cohort Analysis, SIP Continuity Analysis, and Sector Concentration Analysis provided deeper insights into investment risk and investor behavior.

The interactive Power BI dashboard enabled users to explore industry growth, fund performance, investor demographics, and market trends through dynamic visualizations and KPI metrics. The analysis revealed consistent SIP growth, strong investor participation in equity funds, the dominance of SBI Mutual Fund in AUM, and the importance of diversification in reducing investment risk.

Overall, the project demonstrates how data engineering, business intelligence, and financial analytics can be integrated to support data-driven investment decisions and provide valuable insights for investors, analysts, and fund managers.

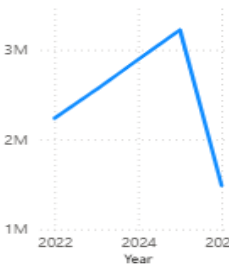
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Sum of nav by Year



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☐ Index

☐ Index/ETF

☐ Large & Mid C...

☐ Large Cap

☐ Liquid

☐ Mid Cap

☐ Short Duration

☐ Small Cap

☐ Value

Industry Overview

Fund Performance

Investor Analytics

SIP & Market Trends

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Industry Overview

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